MILITARY SPECIFICATION

GAGE, PRESSURE, DIAL INDICATING, OXYGEN, HIGH PRESSURE

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 This specification covers one type of high pressure oxygen system pressure gage (range 0-2000 psi) with a 1-3/4 inch round dial and a pointer movement of 180 degrees.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings and publications form a part of this specification. Unless otherwise specified, the issue in effect on date of invitation for bids shall apply.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-N-6011	Nitrogen, Liquid and Gas
MIL-P-6906	Plates, Information and Identification
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery

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SPECIFICATIONS

Military (Continued)

MIL-O-21749	Oxygen, Aviator's Breathing
MIL-L-25142	Luminescent Material, Fluorescent
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of U. S. Military Property
MS33585	Pointers, Dial, Standard Design of Aircraft Instrument
MS33586	Metals - Definition of Dissimilar
MS	Gage, Pressure, Dial Indicating, Oxygen, High Pressure

(When requesting any of the applicable documents, refer to both title and number. All requests should be made via the cognizant Government inspector. Copies of this specification and other unclassified specifications and drawings required by contractors in connection with specific procurement functions should be obtained upon application to the Commanding Officer, Naval Supply Depot (Code CDS), 5801 Tabor Avenue, Philadelphia 20, Pennsylvania. All other documents should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Qualification - The gage furnished under this specification shall be a product which has been tested, and passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 Materials - Materials shall conform to applicable specifications and shall be as specified herein and on applicable drawings. Materials which are not covered by specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.2.1 Metal parts - All metal parts shall be of a corrosion resistant material or treated in a manner to render them adequately resistant to corrosion.

3.2.1.1 Dissimilar metals - Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in Standard MS33586.

3.2.2 Nonmagnetic materials - Nonmagnetic materials shall be used for all parts of the instruments.

3.2.3 Luminescent materials - The fluorescent luminescent material specified shall be in accordance with Specification MIL-L-25142.

3.2.4 Protective treatment - When materials are used in the construction of the gage that are subject to deterioration when exposed to environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. Protective coating which might crack, chip, or scale during normal service life or under extremes of environmental conditions shall not be used.

3.3 Design and construction - The design and construction of the gage shall be in accordance with Standard MS , or as specified in the contract or order (see 6.2).

3.3.1 Operating mechanism - The indicating and the pressure elements shall comprise a unit distinct from the case.

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3.3.1.1 Overpressure stop - An overpressure stop shall be provided on the gage designed so as to restrain the primary mechanism rather than the pointer. The overpressure stop shall allow the pointer to travel beyond full scale to 2,200 psi, plus or minus 50 psi. When extreme overpressure occurs, the stop shall not allow the pointer to move beyond the bottom center of the dial.

3.3.1.2 Hair springs - The hair springs shall be made from phosphor bronze.

3.3.1.3 Surge pressure restriction - A restriction of 0.020 inch diameter shall be provided between the gage connection and gage mechanism.

3.3.2 Dial - The dial shall be fastened securely upon the frame of the mechanism by at least two screws in such a manner that it shall not loosen or turn when the gage is vibrated.

3.3.2.1 Dial-cover-glass distance - The distance between the dial and inside of the cover glass shall be the minimum practicable and shall not exceed 0.125 inch.

3.3.3 Pointer - The pointer shall conform to MS33585-1. It shall be light and sufficiently rigid to prevent oscillation under vibration, and shall be free to swing beyond either end of the range without being engaged by a mechanical stop. Suitable stops shall be incorporated within the mechanism to prevent the sector from being disengaged from the pinion. The pointer shall be firmly attached to the mechanism.

3.3.4 Cover glass - The cover glass shall be of the unbreakable or shatter-proof type, shall be clear, flat, and free from flaws that will seriously affect the readability of the gage. The cover glass shall be held in place by a knurled screw on type bezel.

3.4 Performance - The gage shall pass the performance requirements specified in Section 4 when subjected to the applicable tests.

3.5 Interchangeability - All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of Specification MIL-D-70327 shall govern changes in the manufacturer's part numbers.

3.6 Weight - The weight of the gage shall not exceed 3.20 ounces.

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3.7 Operation markings - All markings shall be durable to withstand usage encountered in service.

3.7.1 Numerals - Numerals shall distinctly indicate the graduation to which each applies. When several numerals are used in one group, the space between the numerals shall be approximately 0.015 inch.

3.7.2 Dial and pointer markings - The markings, dimensions, and finishes shall be as specified in Table I.

Table I. Dial and Pointer Markings

Markings	Height or Length (inch ± 0.010)	Width of Line or Graduation (inch ± 0.005)	Material or Finish
Dial			
Numerals "0, 500, 1000, 1500 and 2000"	0.080	0.031	Fluorescent Luminescent
Graduations[1]	0.070	0.031	
Lettering "Oxygen Cylinder Pressure"	0.100	0.015	Fluorescent Luminescent
Arc from 0 to 300 PSI	Width of Arc: 0.31 ± 0.005		
Background of Dial	——	——	Durable Dull[2] Black
Pointer			
Shaded portion	——	——	Fluorescent Luminescent
Unshaded portion	——	——	Durable Dull[2] Black
Lettering "USE NO OIL"	0.080	0.015	Fluorescent Luminescent
Lettering "FULL"	0.080	0.015	Fluorescent Luminescent

1 Graduations shall be evenly spaced. The 1,000 psi graduation shall be located on the vertical centerline.

2 Lusterless black paint, approximately matching color number 37038 of Standard FED-STD-595 shall be used where black paint is specified.

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3.8 Identification of product - The gages shall be marked for identification in accordance with Standard MIL-STD-130. The gages shall be provided with a nameplate conforming to Specification MIL-P-6906.

3.8.1 Warning note - A decalcomania, or other suitable material, marked in red letters (approximately 1/8-inch high) on a white background, containing the following warning note shall be attached to the case:

"USE NO OIL"

3.9 Workmanship - The gages shall be uniform in quality and shall be free from irregularities or defects which could adversely affect performance, reliability, or durability. The gages shall be free of oil, grease, fuel, water, dust, dirt, or any other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of examination and tests - The examination and testing of the gages shall be classified as follows:

(a) Qualification inspections

(b) Preproduction inspections

(c) Quality conformance inspections

4.3 Qualification inspections - Qualification inspections shall consist of all the inspections of this specification.

4.3.1 Sampling for qualification inspections - Qualification samples shall be accompanied by the manufacturer's inspection reports. Qualification samples shall consist of three gages. Samples shall be forwarded to the Supply Officer,

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Naval Air Material Center, Philadelphia 12, Pennsylvania, Attention: Director, Air Crew Equipment Laboratory. The samples shall be plainly identified by securely attached durable tags marked with the following information:

> Sample submitted by (name) (date) for Qualification inspections in accordance with the requirements of Specification MIL-G- (Wep), under authorization (reference Bureau of Naval Weapons letter authorizing the inspection).

4.3.1.1 Manufacturer's inspection report - Prior to forwarding qualification samples to the inspecting activity, additional samples shall have been tested by the manufacturer, or an approved inspecting facility, in accordance with this specification. The report shall include the results of all tests including a detailed statement indicating compliance or extent of non-compliance with each requirement of this specification, referring specifically to the applicable paragraph numbers in this specification.

4.4 Preproduction inspections - Unless otherwise specified, as soon as practicable after award of a contract or order, the contractor shall furnish preproduction samples for inspections to determine conformance with this specification. The number of samples and the forwarding of the samples shall be as specified in 4.3.1 except that the contract number and manufacturer's part number shall be included. Preproduction inspections may include any or all of the inspections of this specification. Quantity production shall be withheld until the preproduction sample has been pronounced satisfactory by the Government. Preproduction inspections may be waived at the discretion of the Government. The approval of the preproduction samples or the waiving of the preproduction inspections shall not relieve the contractor of his obligation to submit samples for quality conformance inspections.

4.5 Quality conformance inspections - Quality conformance inspections shall consist of individual inspections and sampling tests.

4.5.1 Individual inspections - Each gage on the contract or order shall be subjected to the following inspections. Any gage containing a defect shall be rejected.

(a)	Examination	(4.7.1)
(b)	Scale error at room temperature	(4.7.2)
(c)	Friction	(4.7.3)

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4.5.2 Sampling tests - A random sample shall be selected from each inspection lot in accordance with Standard MIL-STD-105 at inspection level L-6. Failure of any sample to pass the following tests shall be cause for rejecting the lot:

(a) Vibration (4.7.6)

(b) Seasoning (4.7.7)

(c) Drift (4.7.8)

(d) Overpressure (4.7.9)

4.5.2.1 Inspection lot - All gages offered for delivery at one time shall be considered a lot for purposes of inspection.

4.6 Test conditions -

4.6.1 Gas - Unless otherwise specified, the gas used in testing the gage shall be oxygen conforming to Specification MIL-O-21749.

4.6.2 Temperature and pressure - Unless otherwise specified, tests shall be conducted at an ambient temperature of approximately 77° F and at an ambient barometric pressure of approximately 29.92 inches of mercury. If tests are conducted at different ambient conditions, appropriate correction factors shall be applied to the pressure readings. All specified test pressures are gage pressures measured relative to the ambient air pressures.

4.6.3 Vibration - The vibration test shall be conducted with the gage installed on a suitable test fixture, simulating as nearly as practicable an actual installation.

4.7 Inspection methods - The following inspections shall be conducted in the order indicated.

4.7.1 Examination - Each gage shall be thoroughly examined to determine conformance with this specification and applicable standards with respect to all the requirements not covered by tests.

4.7.1.1 The gages shall be examined to ascertain that the preparation for delivery conforms to this specification.

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4.7.2 Scale error at room temperature - The gages shall be tested for scale errors at pressures listed in Table II. The test shall be made by subjecting the gage to the pressure specified to produce these readings, first with pressures increasing, then with pressures decreasing. With the pressures increasing, the pressure shall be brought up to, but shall not exceed the pressure specified to give the desired reading; and with the pressure decreasing, the pressure shall be brought down to, but shall not fall below the pressure specified to give the desired reading. The scale errors at room temperature shall not exceed the tolerances specified in Table II.

Table II. Scale Error at Room Temperature

Test pressure pounds per square inch	Tolerance pounds per square inch
300	± 50
600	± 50
800	± 50
1,000	± 50
1,200	± 50
1,400	± 50
1,600	± 50
1,800	± 50
2,000	± 50

4.7.3 Friction - The gage shall be tested for friction at each alternate test point indicated in Table II, beginning with the second test point. The pressure shall be increased so as to bring the pointer approximately to the desired reading, and then held constant while two readings are taken, the first taken before the gage is tapped, the second taken after the gage is tapped. The difference between any two such readings is the friction error, and shall not exceed the tolerances specified in Table III. This test may be combined with test for scale error at room temperature.

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Table III. Test Requirements

Test	Pounds per square inch
Friction test:	
Tolerance	50
Seasoning test:	
Test pressure	2,000
Tolerance	30
Drift test:	
Test pressure	2,000
Tolerance	30
Overpressure test:	
Test pressure	3,000
Tolerance	50

4.7.4 Low temperature - The gages shall be placed in a chamber and maintained at a temperature of approximately -65° F for a period of 4 hours. At the end of this period and while still at that temperature, the gages shall be tested for operation to determine that the pointers move smoothly throughout the entire range of travel. While at this temperature, the gages shall then be subjected to and pass the scale error at room temperature and friction tests. Not less than 1 hour following this exposure to low temperature, the gages shall be returned to room temperature and subjected to and pass the scale error at room temperature and friction tests. There shall not be any evidence of damage.

4.7.5 High temperature - The gages shall be subjected to a temperature of approximately 160° F for a period of 4 hours. The gages shall then be subjected to and pass the scale error at room temperature test while at this temperature. Not less than 1 hour following this test, the gages shall be returned to room temperature and then be subjected to and pass the scale error at room temperature test. There shall not be any evidence of damage.

4.7.6 Vibration - The gages shall be subjected to vibration for a period of 3 hours. The vibration shall be a circular motion in each plane with respect to the instrument, with a double amplitude of from 0.018 to 0.020 inch diameter and frequency of 300 to 3,000 cycles per minute.

4.7.6.1 After the 3-hour period of vibration and while gage is still being vibrated, an oxygen pressure of 1000 psi shall be applied to the gage. Pointer oscillation and variation shall not exceed ± 50 psi. The gage shall then be subjected to and pass the scale error at room temperature test. There shall not be any evidence of damage.

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4.7.7 Seasoning - The gages shall be subjected to 300 applications at the designated pressure specified in Table III. Not less than 1 hour later, the gage shall then be subjected to and pass the scale error at room temperature test. There shall not be any evidence of damage.

4.7.8 Drift - The gage shall be subjected to the pressures specified in Table III for 1 hour. The increase in reading shall not exceed the tolerance specified in Table III.

4.7.9 Overpressure - The gage shall be subjected to the overpressure specified in Table III for a period of 10 minutes. Not less than 1 hour after the release of the overpressure, the gage shall then be subjected to and pass the scale error at room temperature test. There shall not be any evidence of damage.

4.7.10 Sand and dust - The gages shall be subjected to the sand and dust test specified in Specification MIL-E-5272, Procedure I. The gages shall then be subjected to and pass the scale error at room temperature test. There shall not be any evidence of damage.

4.7.11 Humidity - The gages shall be subjected to the humidity test specified in Specification MIL-E-5272, Procedure I. The scale error at room temperature test shall again be conducted at the completion of the cycles and the change in indication shall not exceed the initial scale error at room temperature test by more than 25 percent. There shall not be any evidence of corrosion of the components. At the completion of the test, the gage shall be examined for collection of moisture in the interior of the case, for corrosion of metal parts, or any other damage. A second sample gage, with case removed, shall be subjected to the humidity cycling test, and inspection for the same features shall be conducted with the exception of collection of moisture within the case. There shall not be any evidence of damage.

4.7.12 Salt spray - The gages shall be subjected to the salt spray test specified in Specification MIL-E-5272, Procedure II. The gages shall then be subjected to and pass the scale error at room temperature test. There shall not be any evidence of corrosion.

5. PREPARATION FOR DELIVERY

5.1 The gages shall be prepared for delivery in accordance with Specification MIL-P-7936. Preservation for Level A shall be in accordance with Specification MIL-P-116, Method 1A, without the use of contact preservatives and without cleaning by immersion in solvent. All openings shall be wrapped or sealed to prevent entry of foreign matter.

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5.1.1 Precautionary marking - The following precautionary marking shall appear on each unit package:

"CAUTION: DO NOT ALLOW CONTAMINANTS OF ANY KIND TO BE USED ON OR ABOUT THE OXYGEN GAGES."

6. NOTES

6.1 Intended use - The high pressure oxygen system pressure gage covered by this specification is intended for use in indicating the breathing oxygen pressure in cylinders.

6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Whether preproduction inspection is required (see 4.4).

(c) Selection of applicable levels of preservation, packaging, and packing required.

(d) Applicable methods of cleaning and preservation.

6.3 With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Chief, Bureau of Naval Weapons, Department of the Navy, Washington 25, D.C.; however, information pertaining to qualification of products may be obtained from the Director, Air Crew Equipment Laboratory, Naval Air Material Center, Philadelphia 12, Pennsylvania.

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SPECIFICATION ANALYSIS SHEET
NAVWEPS FORM 9121/3 (8-62)

FORM APPROVED
BUDGET BUREAU NO. 43-R308

INSTRUCTIONS

This sheet is to be filled out by Government or contractor personnel involved in the use of this specification in procurement of products for ultimate use by the activity shown on the reverse of this sheet.

This sheet is provided for obtaining information which will aid the activity shown in insuring that through use of this specification suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this sheet will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to the activity shown.

SPECIFICATION (No. and abbrev. title)

ORGANIZATION	CITY	STATE

CONTRACT NUMBER	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THIS SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
(If so, give paragraph number and wording, and recommendations for correcting the deficiencies.)

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO IF YES, IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If additional papers are attached, send them fore and papers in an envelope. This form is addressed for use in window envelope when appropriate.)

SUBMITTED BY (Print name and title)	DATE

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